IEOR 145/245: Course Project Information

Project Description and Expectations The main goal of the project is for you to independently investigate a focused topic in revenue management. As described in the syllabus, the use of generative AI tools is *encouraged* for the course project. These tools make it much easier to search the literature, write and debug code, run experiments, and produce well-formatted documents. Correspondingly, **expectations for the depth, scope, and quality of project work are significantly higher this year than in previous offerings of the course**. Strong projects will make modeling choices, defend assumptions, interpret ambiguity, and critically evaluate trade-offs. Projects that primarily consist of literature summaries or AI-generated overviews will not receive high marks.

In previous years, a project could focus on just one of modeling, theory, or experiments. This year, **every project is expected to combine components 1–3 below** (modeling, data and empirics, and some theory), although the balance between them can vary from project to project. In addition, every project will be held to a much higher standard of presentation (component 4).

1. Modeling. Identify a concrete problem, ideally tied to a real application, and build a mathematical model for it to capture the decisions, the sources of uncertainty, the constraints, and the objective. Explain the rationale behind your modeling choices and discuss the assumptions, strengths, and limitations of your model. When applicable, start from models we saw in class and explain clearly what is different about your setting and why the standard models fall short. Once you have a model that you are satisfied with develop solution approaches. Start by developing a basic approach based on techniques introduced in this class before and clearly identify shortcomings of the basic approaches before you explore other (possibly more sophisticated) ones.

2. Data and empirics. Numerical experiments can be great tool for navigating various approaches and understanding shortcomings of the solution techniques and the models themselves. To perform such comparative experiments you need data. You can either find and use a public dataset, or data you collect yourself (for example from public sources, surveys, or controlled experiments, including experiments with LLMs), or (most common) carefully construct the a synthetic dataset yourself (generated with justification that its distributions and structure reflect reality). Use the data to estimate or calibrate your model and to run numerical experiments that compare different methods. Every experiment should be tied to a clear question that you are resolving and methods should be compared on performance as well as practicality. Note that **simulations or experiments cannot by themselves be the main focus** of a project.

3. Some theory. Include at least some mathematical development of your model or approach. For example, you might prove a helpful or insightful structural property of your model or of the optimal policy in your model, establish a performance guarantee (competitive ratio or approximation ratio) for an algorithm in a special case, extend a result from class to your setting, or construct a counterexample that shows limitations of the proposed solution approaches. You are not expected to prove deep new theorems, but the analysis should be correct, clearly stated, and insightful.

Presentation and report. There will be an in-class presentation and final report, both of which should be very polished and reader friendly. Write for a reader who knows the course material but not your topic. The report should open with a clear statement of the question you study and a short summary of your main findings. Notation should be defined before it is used and kept consistent. Every figure and table should be legible, with labeled axes, units, and an informative caption. With the tools available

today, typos, unreadable figures, and disorganized writing are not acceptable. At the same time, please keep in mind that polish is about clarity and insightfulness not length. In my experience, long, generic writing with lots of buzz-words (a common feature of unedited AI-generated content) makes a report harder to read and will be penalized. Presentations should be well rehearsed, finish on time, and tell a clear story.

You remain fully responsible for the accuracy, originality, and integrity of everything you submit. Every member of the team should be able to explain and defend any modeling choice, derivation, or piece of code in the project.

Example project ideas We include some example directions below. Each can be scoped in many different ways and each connects to several topics from the course. You are welcome to pursue one of these, but we would be most excited to see you self-investigate and identify problems that genuinely motivate you. All the best projects in memory have been self-motivated. **My main advice is to find your own topic and pick something that you can genuinely enjoy exploring and have fun with!**

- **Advertising inside LLM responses.** LLM-based assistants are quickly becoming a major channel through which people search for information and products, and advertising is starting to appear in them (for example, OpenAI began testing ads in ChatGPT in 2026). In search advertising, an ad occupies one of a few well-defined slots next to the organic results. In an LLM response, an ad may instead be woven into, or placed alongside, generated text, and there is currently no standard practice for how this should be done. We are most interested in how *you* would do it. Innovate on the fundamental models. Start by deciding what exactly is being sold and what are the big new factors to model.

- **Customer choice in the LLM age.** This is actually two different ideas under one bullet. First, a growing number of companies use LLMs to *generate* choice data, for example by prompting a model to act as a "synthetic consumer" who chooses among products at different prices, as a cheaper and faster substitute for surveys and experiments with real customers. How good is such data at capturing real behavior, and what are its limitations? How good are existing choice models at capturing "LLM generated choice"? Second, customers increasingly shop through AI agents that search, compare, and maybe even purchase on their behalf. An agent may consider far more options than a person would, may respond very differently to product position, prices, and descriptions, and may have its own systematic biases. How should the choice of an agent be modeled, and how should firms rethink their assortment and price optimization strategies when some or all customers shop using agents?

- **Resource allocation in data centers.** Demand for AI has made data center capacity (GPUs, memory, power) one of the scarcest and most valuable resources in the economy. Jobs such as training runs, inference requests, and batch computations arrive over time with different resource requirements, durations, priorities, and values, and must be assigned to machines with limited capacity. Formulate basic models for both the offline version of the problem (all jobs are known in advance) and the online version (jobs arrive sequentially and each must be handled on arrival). Adapt algorithms from class to your setting, explore other algorithms , compare them numerically, and analyze their pros and cons, including performance guarantees where possible.

- **Mechanism design for affordable housing in Berkeley.** Demand for affordable (below market rate) housing in Berkeley vastly exceeds supply. By design these units are allocated through lotteries (instead of prices), often combined with priority rules. Look into how these lotteries work in practice (the City of Berkeley and Berkeley Housing Authority websites are good places to start). What are the bottlenecks? How would you model these bottlenecks and improve the existing mechanisms? How do current practices relate to mechanisms in the literature? Propose solutions, evaluate them

against the status quo in terms of efficiency, fairness, and incentives (for example, with simulations calibrated to publicly available information), and discuss the barriers to adopting them in practice. Note that this direction draws on mechanism design without money, which we will cover at the end of the course if time permits, so you would need to read ahead.

Sources and references The material for your project can come from anywhere, as long as it involves some aspect of revenue management or uses revenue management concepts. Some suggestions for inspiration:

- **Journals and conferences:** Operations Research, Management Science, Manufacturing & Service Operations Management, INFORMS Journal on Applied Analytics, as well as conferences at the intersection of computer science and economics such as the ACM Conference on Economics and Computation (EC) and The Web Conference (WWW).
- **News articles:** From The Economist, Harvard Business Review, Reuters, New York Times, Wall Street Journal (UC Berkeley students have free access to the last), etc.
- **Data:** Good public data can be hard to come by so start looking early. Some examples of public datasets: cluster traces (Google, Alibaba, Azure), hotel search and booking data (Expedia, on Kaggle), retail transaction data, and ad click logs (e.g., Criteo).

Logistics and submission deadlines:

- The project must be done in teams of 4-5 students. In general, teams with less than 4 students are not allowed unless you have a very good reason and get prior approval from the teaching team. Teams with more than 5 students are not allowed – no exceptions. Students in 145 can team up with students in 245. Ideally, your project team should also be your team for the group simulation exercise.
- To facilitate team formation, a google sheet will be shared alongside this document. Please enter your name and other details on the sheet and use it to reach out to other potential teammates.
- Recall that the project is worth 55% of your final grade: 25% for the presentation, 25% for the final report, and 5% for timely completion of Milestones 1 and 2 below. The project will also include a peer scoring component to encourage all team members to contribute (details to follow).
- **Milestone 1: Final team information.** Please form your team by 11:59 pm pacific time on October 06. An assignment will be set up on bCourses with instructions for submitting your team. Timely submission counts toward the milestone component of your grade. Please do not add yourself to an existing team without communicating with them.
- **Milestone 2: Project topic and abstract.** A short project abstract will be due by email at 11:59 pm pacific time on October 20. In one paragraph, describe what you are planning to do so that we can offer you feedback if we feel that you may be on the wrong track. Please read this document carefully to understand the expectations from your project. Send one email per team to both rudwani@berkeley.edu and suho_kang@berkeley.edu, and cc all your teammates. Timely submission counts toward the milestone component of your grade.
- **Milestone 3: Project presentation.** Each team will give a live 10-15 min presentation of their preliminary findings during the week of November 30, the last week of classes. The time limit will be finalized after we know the total number of teams. Presentations cannot be prerecorded.

- **Milestone 4: Project report.** The project report will be due on bCourses at 11:59 pm pacific time on December 14. The gap between the presentations and the report due date is designed to provide some time to refine your work based on our feedback after you present. Each report can be up to 6 pages, including figures but excluding references. Reports should be typeset with 11pt font and 1 inch margins. Only one member from each team should upload the report. Further details will be announced later on.

- To get a concrete idea of the format and scope of a project, please take a look at the sample reports from prior years that have been uploaded along with this document. Given the higher expectations this year, treat these samples as a baseline to exceed rather than a target. Goes without saying: your project topic must be different from the uploaded samples.

- Please note that **cross-course duplication is strictly prohibited**, i.e., submitting a project that is used (or has been used) for another course will result in a grade of zero and a report to the Center for Student Conduct. If you are unsure whether your project overlaps too much with work you are doing for another course, please check with the teaching team.